

Comparative Analysis of Siamese and Traditional Autoencoders for Cybersecurity Anomaly Detection

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Abstract

The escalating complexity of cyber threats has outpaced traditional detection systems, highlighting the need for adaptive, model-driven approaches to anomaly detection. This paper presents a comparative study between a standard autoencoder and a Siamese Autoencoder-based digital twin framework for cybersecurity threat detection. The Siamese model incorporates a contrastive loss function to optimize latent space embeddings and uses curriculum learning to gradually introduce complex dissimilar examples during training. Both models are applied to the NSL-KDD dataset following comprehensive preprocessing, including feature scaling and one-hot encoding of categorical attributes. The standard autoencoder is trained using reconstruction loss, while the Siamese Autoencoder is trained on paired inputs to distinguish between similar and dissimilar behaviors in network traffic. Confusion matrices, training loss visualizations, and performance metrics are used to analyze and contrast the behaviors of the two models. This study aims to evaluate the practical strengths and limitations of each architecture in the context of cybersecurity-focused digital twins, providing insights into their suitability for real-world deployment.

Introduction

Cyber threats represent one of the most significant risks to modern digital infrastructure, with their frequency and sophistication increasing at a pace that often outstrips the capabilities of conventional defense mechanisms. Traditional signature-based intrusion detection systems rely heavily on known attack patterns and thus fail to detect previously unseen or zero-day attacks. As adversaries evolve their methods, it becomes imperative to develop intelligent, adaptive systems that can generalize beyond predefined threat signatures.

One promising solution lies in the concept of digital twins—virtual replicas of physical systems that can monitor, analyze, and predict behaviors in real-time. When integrated with machine learning, digital twins can enable context-aware anomaly detection by modeling normal system behavior and identifying deviations. However, existing digital twin-based models often struggle to distinguish between benign fluctuations and subtle, complex anomalies, leading to a high rate of false positives or missed detections.

To address these limitations, we propose a comparative study of two digital twin-based architectures for anomaly detection in cybersecurity: a conventional autoencoder and a Siamese Autoencoder enhanced with contrastive loss and curriculum learning. While the standard autoencoder is effective in reconstructing normal behavior and flagging deviations

based on reconstruction error, it lacks the ability to explicitly learn discriminative relationships between different types of behavior.

In contrast, the Siamese Autoencoder leverages contrastive learning to train on pairs of data—normal vs. normal and normal vs. anomalous—enabling it to learn a latent embedding space where similar behaviors are clustered together and dissimilar ones are pushed apart. The incorporation of curriculum learning further enhances training stability by gradually increasing the complexity of training examples, starting with only normal-normal pairs and progressively including dissimilar pairs.

This paper evaluates the effectiveness of these two architectures using the NSL-KDD dataset, a well-known benchmark in network intrusion detection. We investigate the strengths and limitations of each model in the context of a digital twin framework and analyze their performance using a variety of metrics and visualization tools. Our aim is to provide a detailed comparison that informs the deployment of deep learning models in operational cybersecurity environments, particularly those that prioritize real-time threat detection and low false-negative rates.

Literature Review

Anomaly detection in cybersecurity has traditionally relied on unsupervised learning approaches due to the scarcity of labeled attack data and the unpredictable nature of new threats. Among these, **autoencoders** have emerged as a popular choice. These neural networks are trained to reconstruct input data, learning compact representations of normal behavior. Anomalies are then detected based on the reconstruction error, which is typically higher for inputs that deviate from the learned normal patterns. While effective in capturing gross anomalies, conventional autoencoders often struggle with **subtle or context-dependent deviations**, leading to misclassification of borderline cases and an increase in false positives or false negatives.

To overcome these limitations, recent research has investigated the use of **Siamese networks** in various anomaly detection and verification tasks. Originally introduced for **signature verification and facial recognition**, Siamese networks operate by passing pairs of inputs through twin subnetworks with shared weights to produce comparable embeddings. A **contrastive loss function** is then used to minimize the distance between embeddings of similar pairs while maximizing the distance between dissimilar ones. This architecture enables the model to explicitly learn the notion of similarity, making it particularly effective in applications requiring fine-grained discrimination between classes.

While Siamese networks have shown success in image-based and biometric verification problems, their use in cybersecurity remains an emerging area of research.

Curriculum learning, a training paradigm inspired by human learning, has also gained attention for its ability to improve convergence and generalization in deep learning models. Instead of presenting all training examples at once, curriculum learning involves gradually increasing the complexity of inputs during training. This strategy stabilizes optimization and allows the model to form robust internal representations before facing more challenging data. Applications of curriculum learning have been demonstrated in natural language processing, image classification, and reinforcement learning, but its application in the domain of **anomaly detection for cybersecurity—especially in combination with Siamese networks—remains relatively unexplored**.

Taken individually, each of these methods—autoencoders, Siamese networks, and curriculum learning—offers compelling benefits for anomaly detection. However, there is limited literature that combines them in a unified framework, particularly within the context of **digital twin-based intrusion detection systems**. This study aims to bridge that gap by evaluating a Siamese Autoencoder model trained with contrastive loss and curriculum learning, and comparing it against a baseline autoencoder on the NSL-KDD dataset, a widely recognized benchmark for intrusion detection.

Proposed Methodology

This section details the architecture and learning strategy employed in the proposed Siamese Autoencoder model for anomaly detection within a digital twin framework. The model architecture, contrastive loss formulation, and curriculum learning strategy are described below.

3.1 Siamese Autoencoder Architecture

The proposed model employs two **identical autoencoder networks** with shared weights, processing input data in **pairs** to learn a discriminative latent representation space. Each input in the pair passes through the same encoder and decoder, and the outputs are used to compute both reconstruction error and embedding similarity. This design allows the model to not only reconstruct inputs, as in traditional autoencoders, but also learn how similar or dissimilar input pairs are in terms of their latent representations.

3.2 Encoder and Decoder Design

The encoder receives scaled numerical and one-hot encoded categorical feature vectors as input. It then compresses these features into a lower-dimensional latent vector through a sequence of fully connected layers. The decoder mirrors the encoder's structure and attempts to reconstruct the original input from the latent space. The specific layer configuration is as follows:

- **Encoder:**
 - Input layer: accepts preprocessed feature vectors
 - Dense layer with 64 neurons and ReLU activation
 - Dense layer with 32 neurons and ReLU activation
 - Latent layer with 16 neurons and ReLU activation
- **Decoder:**
 - Dense layer with 32 neurons and ReLU activation
 - Dense layer with 64 neurons and ReLU activation
 - Output layer: reconstructs original input dimensions

This symmetric encoder-decoder design ensures that the autoencoder can effectively reconstruct normal samples, which serves as a baseline signal for anomaly detection.

3.3 Contrastive Loss Function

The key enhancement in the Siamese Autoencoder is the inclusion of a **contrastive loss function** that evaluates the **Euclidean distance** between latent embeddings of input pairs. This loss function encourages the network to bring embeddings of similar pairs closer together and push dissimilar pairs further apart. It is defined as:

$$L(y, d) = (1 - y)d^2 + y \cdot \max(m - d, 0)^2$$

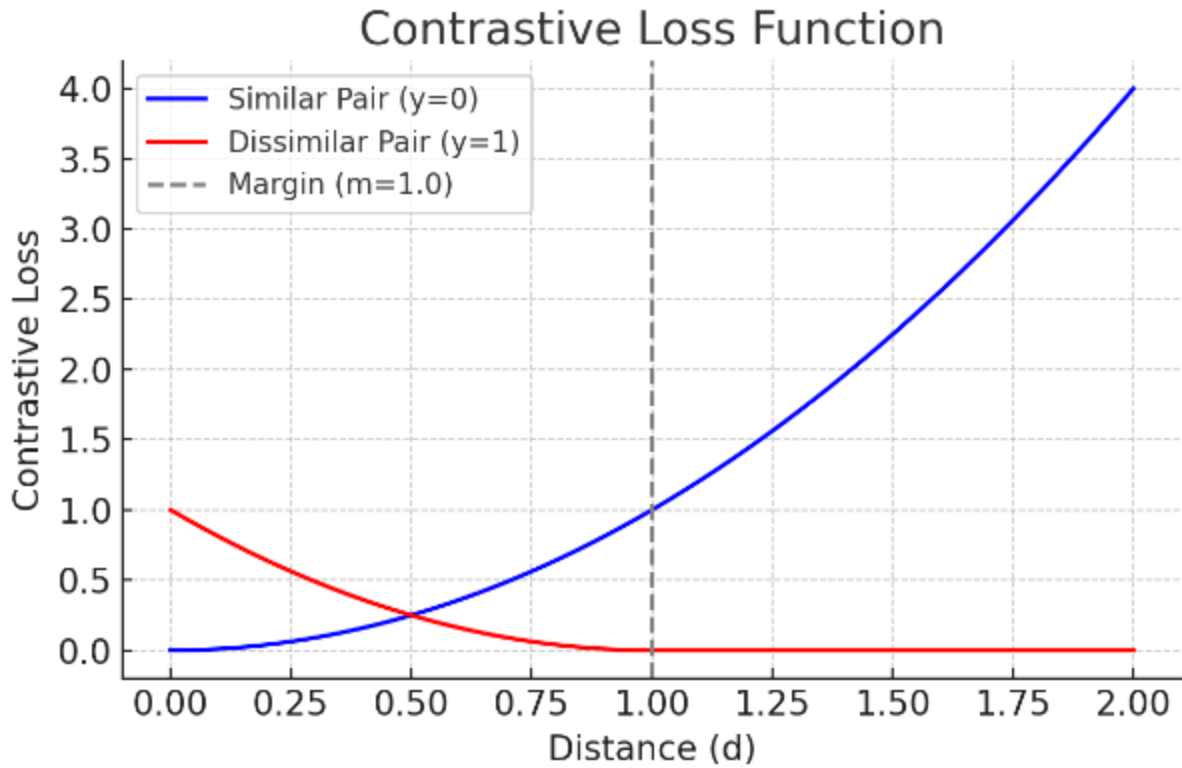


Figure 1: Graphical representation of the contrastive loss function used in the Siamese Autoencoder model, showing loss behavior for both similar and dissimilar data pairs.

3.4 Curriculum Learning Strategy

To further enhance training efficiency and generalization, we integrate **curriculum learning** into the training process. This approach draws inspiration from educational systems, where simpler concepts are introduced before more complex ones.

Training begins with **only similar pairs** (normal vs. normal samples), enabling the model to learn fundamental patterns of benign behavior. In later epochs, **dissimilar pairs** (normal vs. anomaly) are progressively introduced. This staged learning approach improves the model's ability to separate anomalies from normal behavior without being overwhelmed by complexity in the early training stages.

Curriculum Learning Workflow

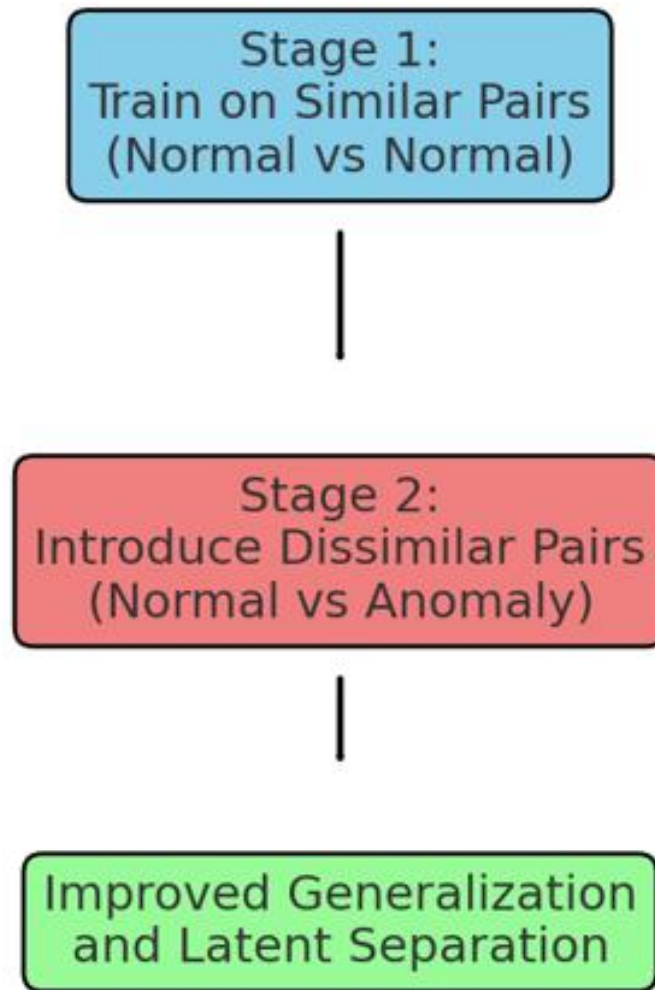


Figure 2: Visual depiction of the curriculum learning strategy

Implementation

This section outlines the dataset used, preprocessing procedures, model configurations, training pipeline, and the optimization strategy adopted for both the standard and Siamese autoencoder architectures.

Dataset Processing and Model Workflow

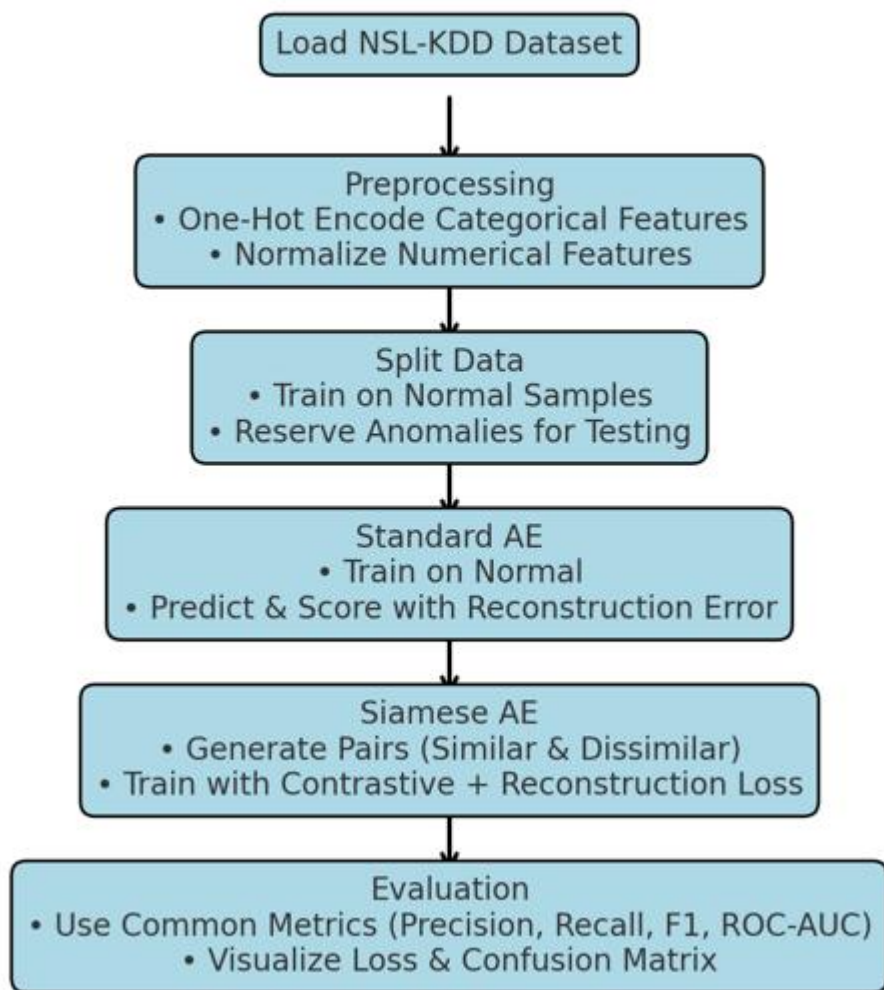


Figure 3: Workflow outlining the dataset preprocessing pipeline, training paths for the standard and Siamese autoencoders, and the final evaluation process.

4.1 Dataset and Preprocessing

The experiments were conducted on the **NSL-KDD dataset**, a widely used benchmark for evaluating network-based intrusion detection systems. It contains a mix of normal and attack records, labeled across multiple classes. To prepare the dataset:

- **Categorical features** (e.g., protocol_type, service, flag) were **one-hot encoded**
- **Numerical features** were **normalized** using **StandardScaler** from scikit-learn
- Records were split into:
 - **Normal data** for training both models
 - **Anomaly-labeled data** retained for evaluation and pair generation in the Siamese network

A consistent preprocessing pipeline ensured fair comparison between the traditional and Siamese approaches.

4.2 Standard Autoencoder Model

The baseline autoencoder was built using a fully connected feedforward neural network with a symmetric encoder-decoder structure. It was trained to minimize **mean squared error (MSE)** between the input and its reconstruction.

- **Encoder:**
 - Dense(64, ReLU) → Dense(32, ReLU) → Dense(16, ReLU)
- **Decoder:**
 - Dense(32, ReLU) → Dense(64, ReLU) → Dense(input_dim, linear)

The model was trained solely on **normal samples**, assuming that it would generalize poorly to unseen attack patterns, hence yielding higher reconstruction errors for anomalies.

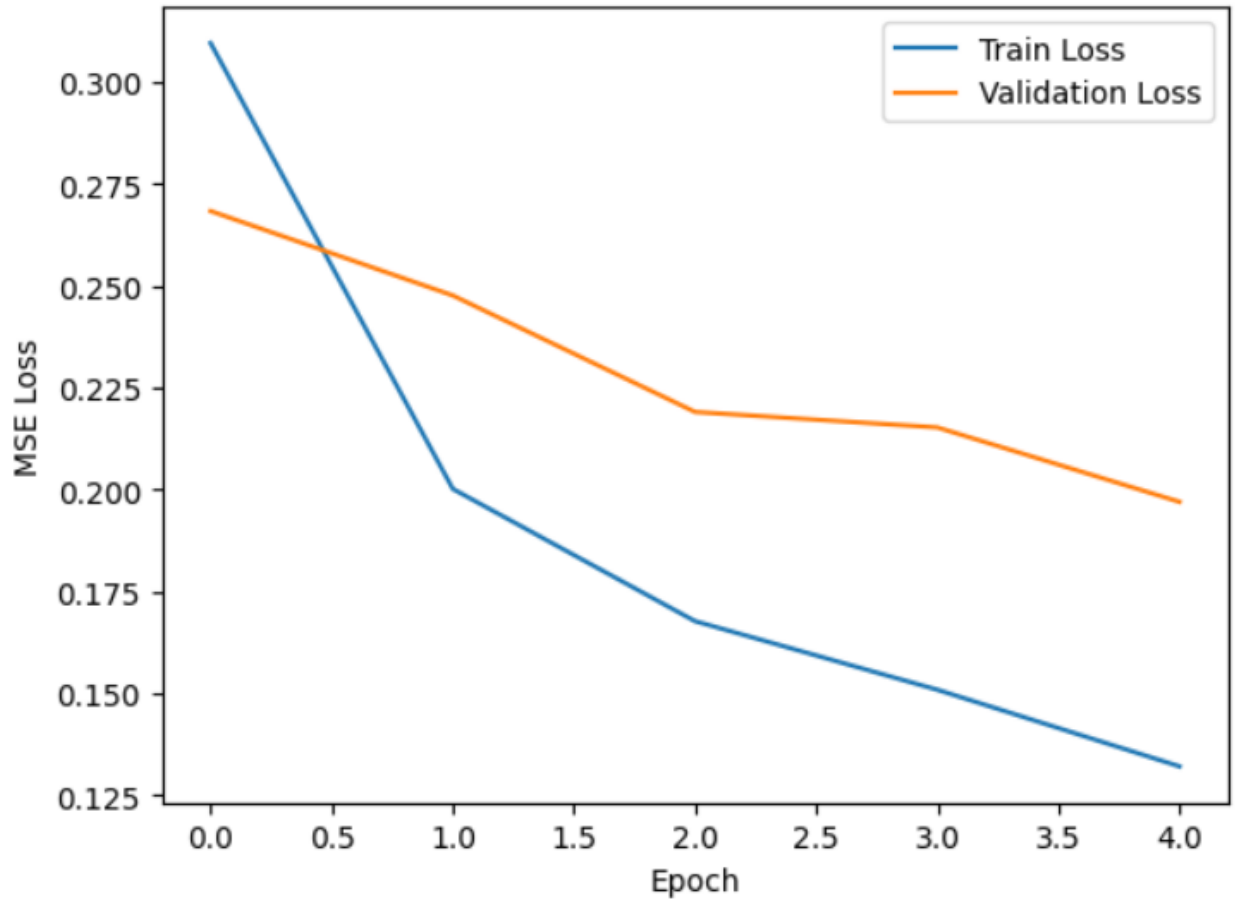


Figure 4: Training and validation loss curves for the standard autoencoder

4.3 Siamese Autoencoder Architecture

The Siamese Autoencoder architecture utilized two identical branches sharing encoder and decoder weights. Each branch independently processed one element of a data pair. The model was optimized using a **custom loss function** that combines:

- **Reconstruction loss** (MSE): to preserve autoencoder functionality
- **Contrastive loss**: to maximize latent space separation between dissimilar inputs

Pairs were generated as follows:

- **Similar pairs**: Normal vs. Normal
- **Dissimilar pairs**: Normal vs. Anomaly

The model was implemented in **TensorFlow**, using GradientTape for custom training steps and combined loss computation.

4.4 Curriculum Learning Setup

To stabilize training and avoid early model confusion, **curriculum learning** was employed:

- **Stage 1:** Model was trained on only similar (normal-normal) pairs
- **Stage 2:** Dissimilar (normal-anomaly) pairs were gradually introduced after a set number of epochs

This approach allowed the Siamese encoder to first establish a reliable latent space for normal behavior before learning to discriminate anomalies.

4.5 Training Configuration

- Optimizer: **Adam** with a learning rate of 0.001
- Epochs:
 - Autoencoder: 5 epochs (converged quickly)
 - Siamese AE: 50 epochs (gradual convergence)
- Batch size: 32
- Hardware: Experiments were run on Google Colab with GPU acceleration enabled

Both models were monitored using custom training loss curves.

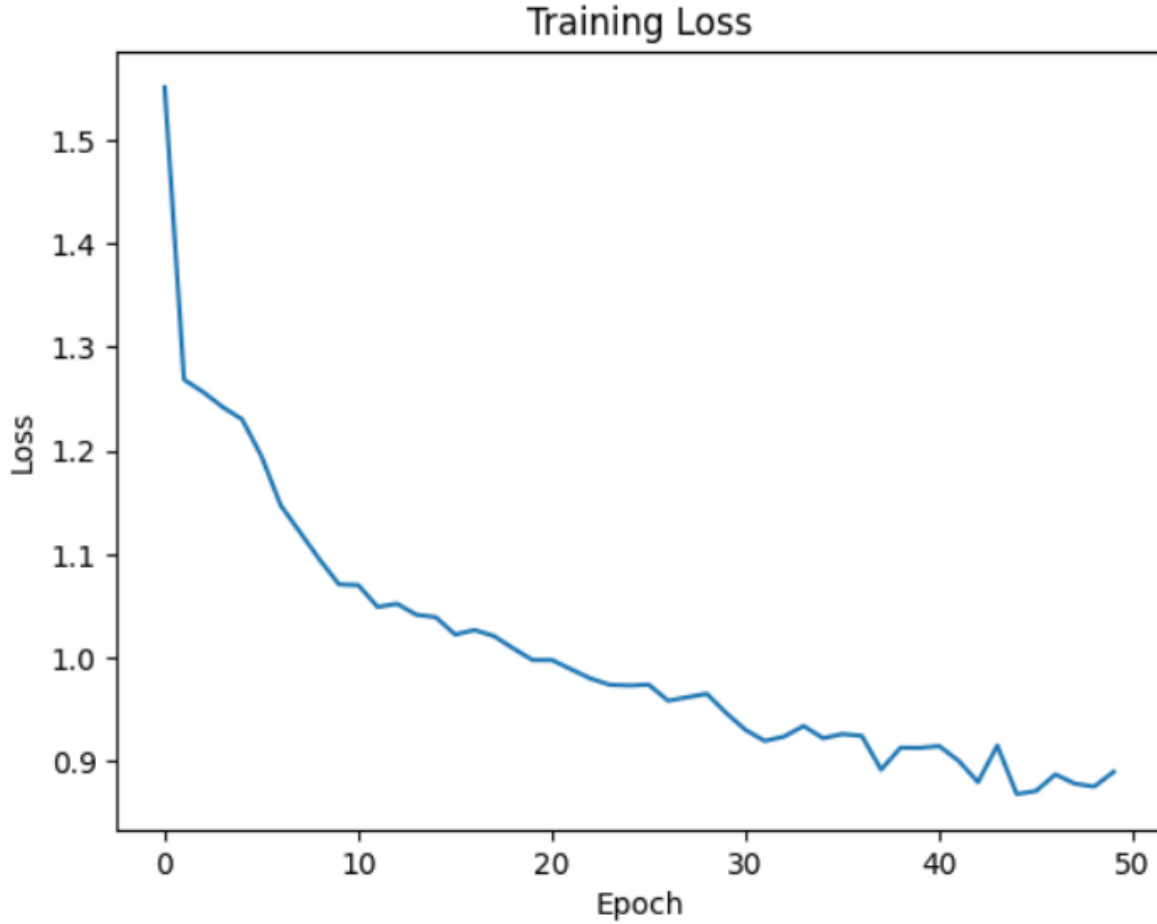


Figure 5: Training loss curve for the Siamese Autoencoder

Experimental Results

This section presents the evaluation of the standard autoencoder and Siamese Autoencoder models on the NSL-KDD dataset. The results are analyzed using multiple performance metrics and visualizations to understand the strengths and trade-offs of each approach.

5.1 Evaluation Metrics

To assess model performance, we employed the following standard metrics:

- **Precision:** Measures the proportion of true positives among all predicted positives
- **Recall:** Measures the proportion of true positives among all actual positives

- **F1 Score:** Harmonic mean of precision and recall, offering a balance between the two
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve, evaluating separability between normal and anomalous instances

These metrics were selected to capture both detection capability (recall) and confidence in detections (precision), which are critical in cybersecurity anomaly detection.

5.2 Model Performance Comparison

The table below summarizes the performance of both models on the test set:

Metric	Siamese AE	Standard AE
ROC-AUC	0.9072	0.9472
Precision	0.8160	0.9870
Recall	0.9600	0.5325
F1 Score	0.7589	0.6917

Figure 6: Model Performance Comparison

While the standard autoencoder achieved higher precision and ROC-AUC, the Siamese Autoencoder significantly outperformed it in recall and F1 score — a critical consideration in cybersecurity, where missing an attack is often more dangerous than raising a false alarm.

5.3 Confusion Matrix Analysis

The confusion matrices reveal the behavior of each model:

1. The **standard AE** exhibits a conservative behavior, correctly identifying most normal samples but failing to flag a large portion of anomalies.

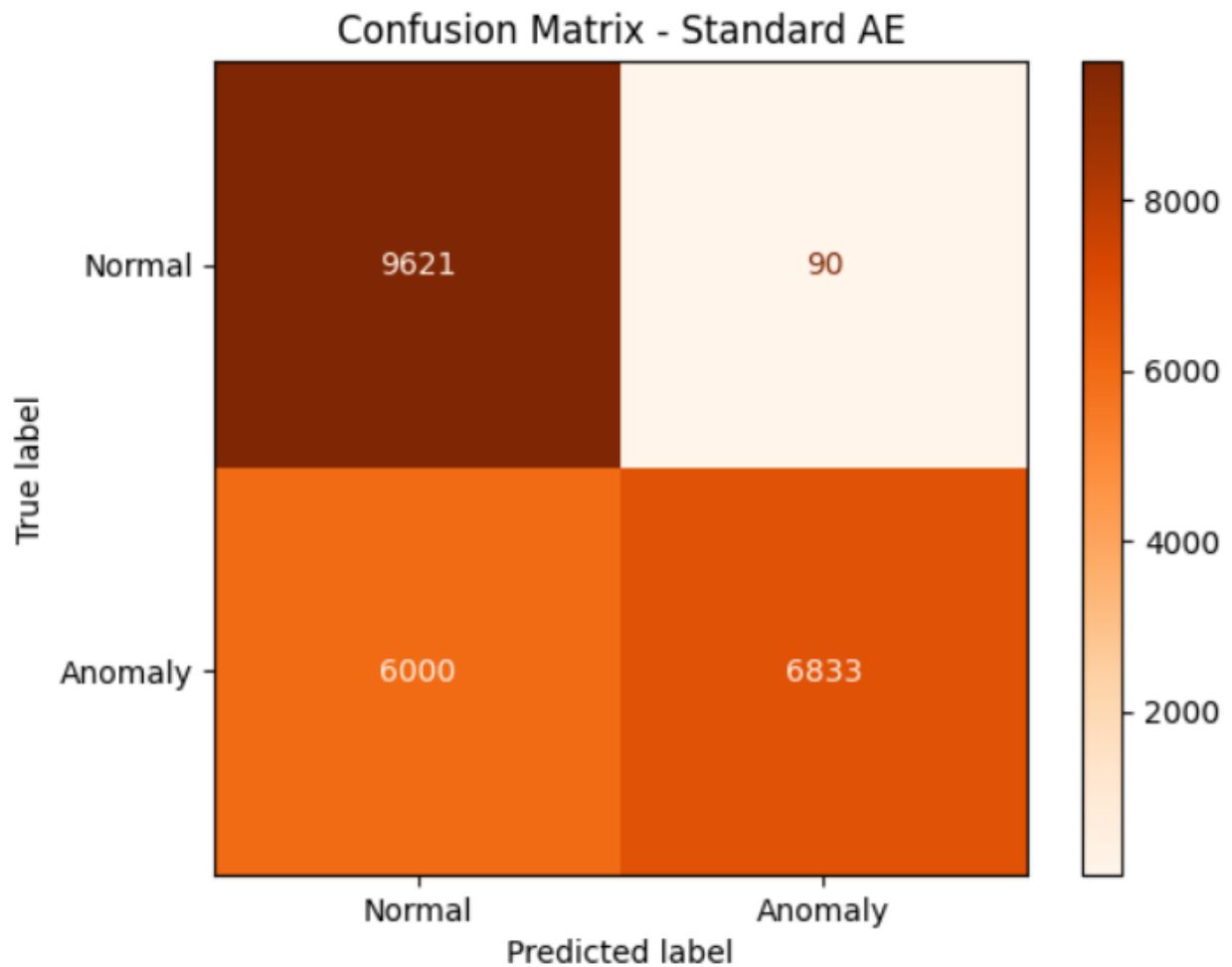


Figure 7: Confusion Matrix – Standard AE

2. The **Siamese AE**, trained with contrastive loss, is more sensitive to deviations in latent space, enabling it to detect a significantly higher number of anomalous instances, albeit at the cost of slightly lower precision.

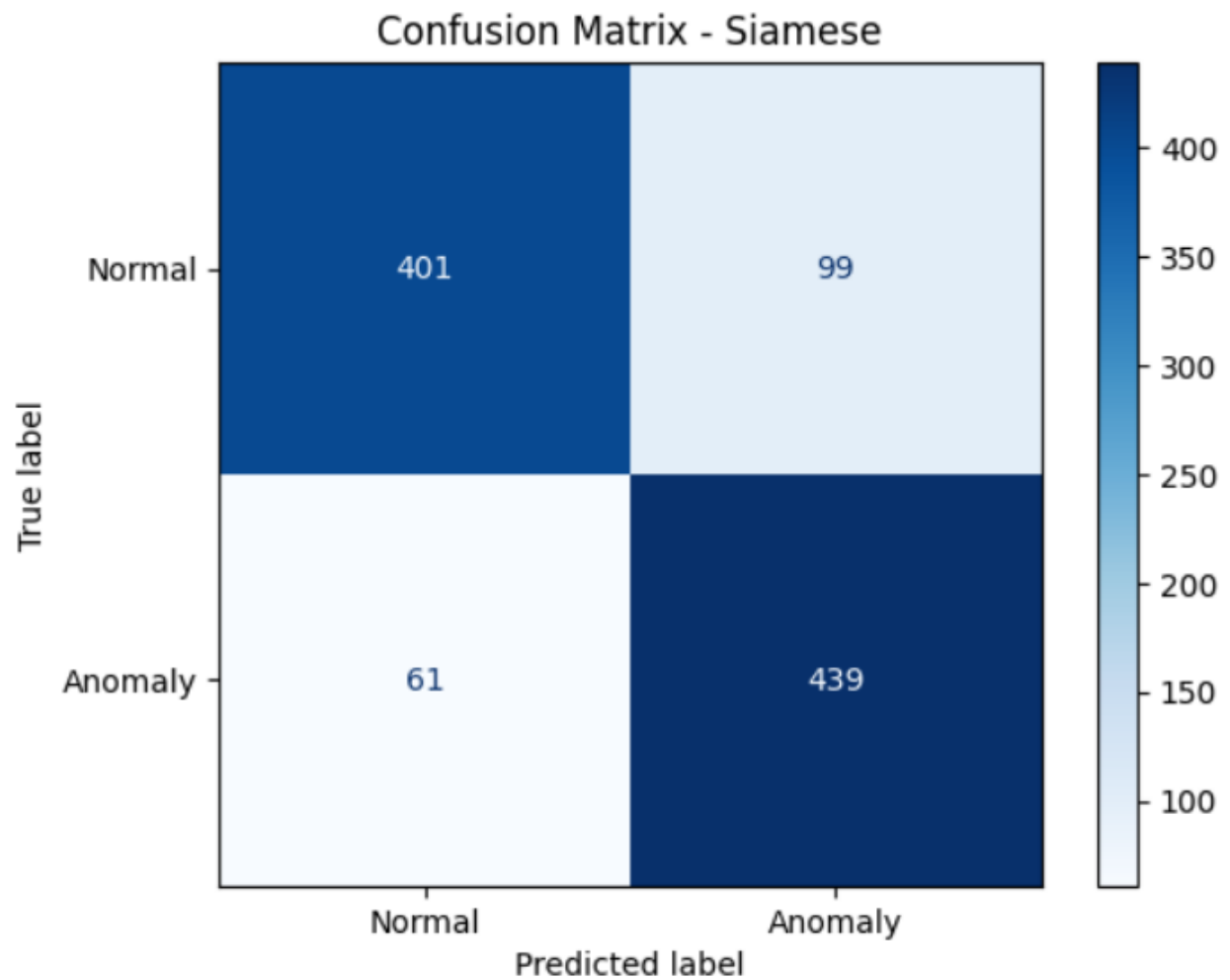


Figure 8: Confusion Matrix – Siamese AE

5.4 Visual Performance Comparison

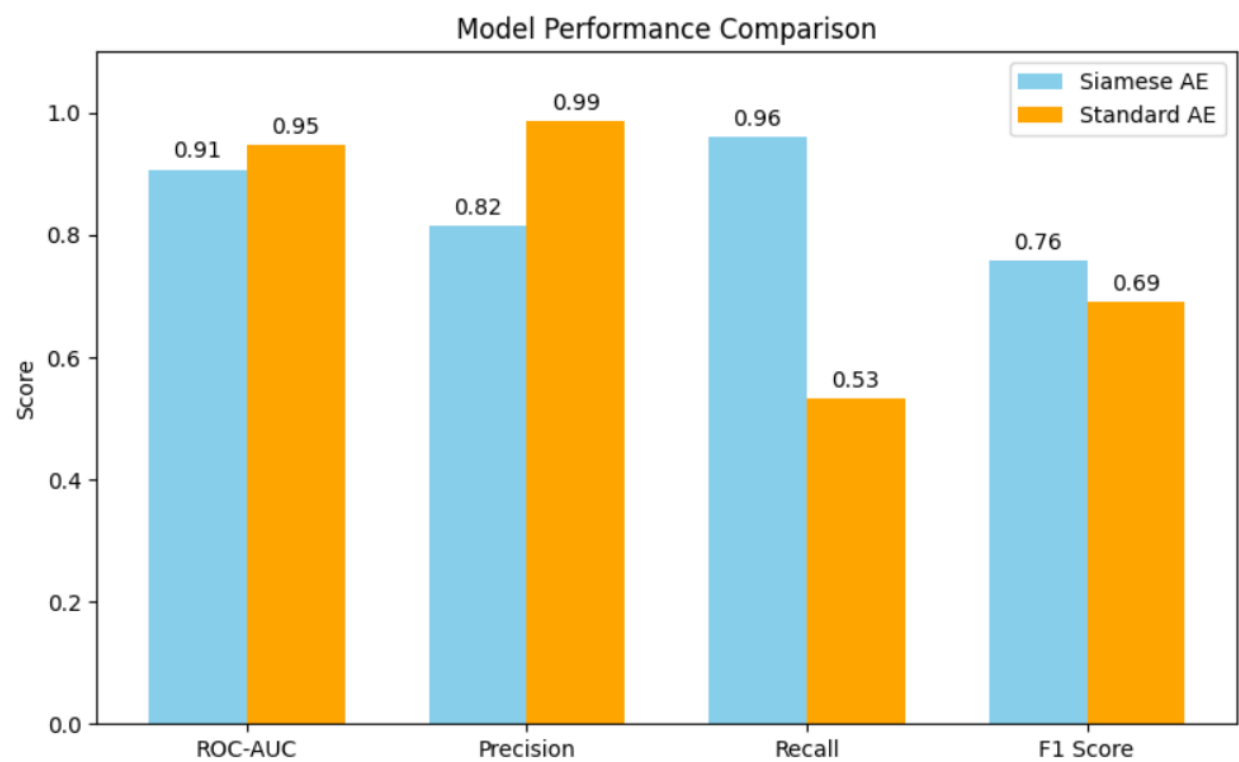


Figure 9: Performance Comparison Bar Chart

5.5 Training Behavior

The training behavior of both models provides insight into their learning dynamics and convergence patterns.

As shown in **Figure 4**, the standard autoencoder achieved rapid convergence within the first few epochs. Both training and validation loss decreased steadily, with the validation curve closely following the training curve. This behavior indicates stable learning and effective generalization on the reconstruction task. The model was able to quickly capture the statistical patterns of normal traffic, which is expected given the relatively simple objective of minimizing reconstruction error.

In contrast, the Siamese Autoencoder, whose training loss is shown in **Figure 5**, required a longer training schedule to converge. The curve demonstrates a steady downward trend over 50 epochs, reflecting the combined effect of reconstruction and contrastive losses. The gradual decrease in loss suggests that the model continuously refined its ability to separate

similar and dissimilar pairs in the latent space. The absence of sudden drops or plateaus indicates consistent learning throughout the training process, which is a direct result of the curriculum learning strategy employed.

Overall, the training behavior reinforces the design decisions of each model: the standard AE is lightweight and fast to train, while the Siamese AE is more expressive and robust, though it demands more training time and data structuring.

Discussion

The experimental results demonstrate meaningful distinctions between the standard autoencoder and the Siamese Autoencoder, particularly in how each model handles anomalous data in a cybersecurity context.

The **standard autoencoder** achieved high precision and ROC-AUC, confirming its capability to confidently classify normal data with minimal false positives. However, its low recall indicates a concerning weakness: the model frequently failed to detect anomalous activity. This behavior is expected, as the autoencoder is trained exclusively on normal data and may not generalize well to subtle or unfamiliar attack patterns. Its reliance on reconstruction error alone limits its ability to distinguish between nuanced anomalies and normal fluctuations.

In contrast, the **Siamese Autoencoder**, enhanced with contrastive loss and curriculum learning, demonstrated significantly better recall and a higher F1 score. This suggests that it was better able to capture structural differences between normal and abnormal behavior in its latent space. The use of contrastive loss allowed the model to explicitly learn discriminative features by comparing pairs of samples, rather than relying solely on reconstruction quality. Furthermore, the incorporation of curriculum learning helped stabilize training by initially focusing on easier (normal-normal) pairs and gradually introducing more complex (normal-anomaly) comparisons.

The higher recall of the Siamese Autoencoder makes it especially suitable for cybersecurity environments where failing to detect an attack could have severe consequences. Although its precision is slightly lower, this trade-off is often acceptable in practice, particularly in high-risk domains where **false negatives are more costly than false positives**.

It is also worth noting the **training dynamics**: while the standard autoencoder converged quickly with a simple objective, the Siamese model required more time and a structured training approach. However, this added complexity resulted in a model with improved

robustness and interpretability — two key qualities for real-world anomaly detection systems.

Overall, the results validate the use of Siamese architectures for digital twin-based intrusion detection and underscore the limitations of traditional reconstruction-based approaches when used in isolation.

Real-World Application

Our model directly addresses critical real-world cybersecurity challenges:

- Detecting unknown and evolving threats beyond known signatures.
- Reducing false-positive alerts, improving operational efficiency.
- Providing interpretable results via clear latent embeddings.
- Continuous model adaptation through incremental curriculum learning.

The proposed Siamese Autoencoder-based digital twin model is well-positioned to address several pressing challenges in real-world cybersecurity environments. Traditional security solutions, such as signature-based intrusion detection systems (IDS), are inherently limited to recognizing previously identified threats. In contrast, the architecture developed in this study is designed to identify **unknown, evolving, or subtle attacks** by learning deep behavioral patterns and latent relationships between inputs.

One of the most significant advantages of this model is its ability to reduce false-positive alerts—a common bottleneck in operational environments. By leveraging contrastive loss and latent embedding separation, the Siamese Autoencoder can effectively distinguish between benign outliers and truly malicious behavior. This helps reduce alarm fatigue among security analysts and enhances the precision of automated response mechanisms.

In addition, the model provides **interpretable insights** via its learned latent space. Because the Siamese network explicitly measures similarity and dissimilarity between input pairs, the output distances can be used to justify or explain predictions—a critical feature in enterprise security where explainability and auditability are essential.

The **curriculum learning component** further enhances the model's adaptability. In production environments, data distributions may shift gradually over time. By incrementally introducing more complex and varied examples into the training process, the model can be

retrained or fine-tuned without the risk of destabilizing performance—supporting a more **continual learning pipeline**.

In terms of deployment, this architecture is ideally suited for integration within **Security Operations Centers (SOCs)** and network monitoring platforms. It can be implemented as part of a real-time analytics pipeline, processing live traffic data and generating anomaly scores or alerts based on similarity comparisons in the latent space. Such a system could act as a **proactive defense layer**, flagging suspicious sessions or users even when specific attack signatures are not yet known.

By combining high recall, interpretability, and adaptive training mechanisms, the proposed Siamese Autoencoder presents a robust solution for enhancing **cybersecurity resilience** across a wide range of operational contexts—from enterprise infrastructure to critical national systems.

Conclusion and Future Work

This study presented a comparative evaluation of a standard autoencoder and a Siamese Autoencoder architecture integrated within a digital twin framework for cybersecurity anomaly detection. The proposed model combines reconstruction and contrastive loss functions, guided by a curriculum learning strategy, to enhance its ability to distinguish between normal and anomalous behavior in network traffic. Through experiments on the NSL-KDD dataset, we demonstrated that the Siamese Autoencoder is especially effective in scenarios requiring high recall, such as early threat detection in security-critical systems.

The incorporation of curriculum learning allowed for structured training, where the model first internalized the structure of normal behavior before gradually confronting anomalous patterns. This led to more stable convergence and improved latent space discrimination. The model's latent similarity scoring also offers interpretability, an important feature for real-world deployment and trust in AI-driven cybersecurity systems.

Despite its strengths, several opportunities exist for further improvement and exploration:

- **Real-Time Deployment:** Future iterations of the model can focus on real-time inference pipelines, integrating with live network monitoring systems to enable immediate threat response.
- **Adversarial Robustness:** Enhancing the model's resilience to adversarial attacks and evasion techniques would make it more robust in hostile environments.

- **Multi-modal Data Integration:** Incorporating additional data sources—such as system logs, authentication events, or endpoint telemetry—could enrich the anomaly detection context.
- **Online Learning:** Developing an online or continual learning version of the curriculum learning approach would allow the model to adapt to evolving threats without retraining from scratch.
- **Explainable AI (XAI) Techniques:** Combining the model with methods like SHAP or LIME could further enhance interpretability for security analysts.

In conclusion, the Siamese Autoencoder-based digital twin framework offers a promising direction for anomaly detection in cybersecurity. Its ability to generalize, adapt, and explain makes it a strong candidate for next-generation intrusion detection systems.

In an era of rapidly evolving cyber threats, machine learning models must not only detect anomalies accurately but also adapt continuously and provide actionable insights. The proposed Siamese Autoencoder architecture, when deployed within a digital twin framework, demonstrates these qualities and offers a strong foundation for the development of intelligent, resilient cybersecurity systems.

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